

Let's get personal: Defining and estimating effects of dynamic treatment strategies using real-world data

ISPE Annual Meeting
August 30, 2026
2:30 PM – 6:00 PM
Allian Milan Convention Center, Milan, Italy

Instructors



Emma McGee, PhD

Research Fellow, Harvard T.H. Chan School of Public Health
Email: emcgee@hsph.harvard.edu



Lucia Petito, PhD

Associate Professor, Northwestern University
Email: lucia.petito@northwestern.edu



Rienna Russo, PhD

Research Fellow, Harvard T.H. Chan School of Public Health
Email: riennarusso@hsph.harvard.edu

Description

Questions about dynamic (i.e., ‘personalized’) treatment strategies that account for evolving individual characteristics increasingly outnumber the randomized trials available to answer them. When trials do not exist, real-world data, e.g., electronic health records and health insurance claims, can be used to estimate the observational analogues of the per-protocol effects of dynamic strategies, though, sufficient data and appropriate methods are required. In this course, participants will learn how to 1) define dynamic treatment strategies, and 2) estimate the effects of dynamic treatment strategies using g-methods in real-world observational datasets using a target trial emulation approach. We will provide practical guidance, including an overview of data requirements for the estimation of the effects of dynamic strategies and a theoretical review of key aspects of dynamic treatment strategies. Participants will gain experience applying concepts to real-world data during hands-on sessions, with R and SAS code provided.

Learning Objectives

Upon successful completion of this course, you should be able to:

1. Define dynamic treatment strategies and motivate their use in personalized medicine
 2. Formulate well-defined causal questions about dynamic treatment strategies via the specification and emulation of a target trial
 3. Understand how to use real-world observational data and g-methods to estimate the effects of dynamic treatment strategies and avoid common pitfalls
 4. Perform practical data analyses implementing these methods in R and SAS
-

Structure and Schedule

The course will be structured as a series of lectures and exercises. Lectures will introduce core theoretical concepts. Exercises will provide the opportunity to apply these concepts to real-world data through target trial emulation and statistical analyses implemented in R and SAS. The schedule is described below:

Time	Activity	Objectives
2:30 – 3:15 pm	Lecture 1 Defining dynamic treatment strategies	• Motivate the use of dynamic treatment strategies in personalized medicine • Define dynamic treatment strategies • Review common pitfalls • Distinguish between different causal estimands
3:15 – 4:05 PM	Exercise 1 Defining dynamic treatment strategies	• Use target trial emulation as a tool for defining dynamic treatment strategies • Discuss a dynamic pharmacoepidemiologic research question in the context of the specification and emulation of a target trial
Coffee break		
4:15 – 5:00 PM	Lecture 2 Estimating the effects of dynamic treatment strategies	• Understand how to estimate the effects of dynamic strategies using inverse probability weighting • Review key assumptions • Describe data requirements • Identify alternative estimators
5:00 – 5:50 PM	Exercise 2 Estimating the effects of dynamic treatment strategies	• Implement appropriate causal analyses of dynamic treatment strategies in R and SAS using inverse probability weighting • Generate and interpret results
5:50 – 6:00 PM	Wrap-up	• Wrap-up and Q&A

Course Materials

All course materials are available via GitHub: <https://github.com/emma-mcgee/dynamic-treatment-strategies>

Expectations

Course participants are expected to attend all sessions and actively participate in discussions. Participants are strongly encouraged to ask questions and to work collaboratively in small groups during the exercise sessions.

Pre-requisites

This course is well-suited to individuals in the field of pharmacoepidemiology with an intermediate knowledge of causal inference techniques. Participants are expected to have:

1. Experience with the analysis of health data
2. Prior introductory courses on epidemiologic study design and data analysis
3. Working knowledge of R or SAS, with either R (version 3.4 onwards) or SAS (version 9.4 onwards) installed on their computers prior to the start of the course
4. Participants unfamiliar with the target trial emulation approach should read PMID: 26994063
5. Participants with no prior experience with inverse probability (IP) weighting should watch a series of videos on both nonstabilized and stabilized IP weights (freely available at <https://youtu.be/Uyb5U34EIwA>, and <https://youtu.be/O3R9l0G0Q9I>) and read Chapters 12, 21.2 and 21.5 of the “Causal Inference: What If” book (freely available at <https://miguelhernan.org/whatifbook>)